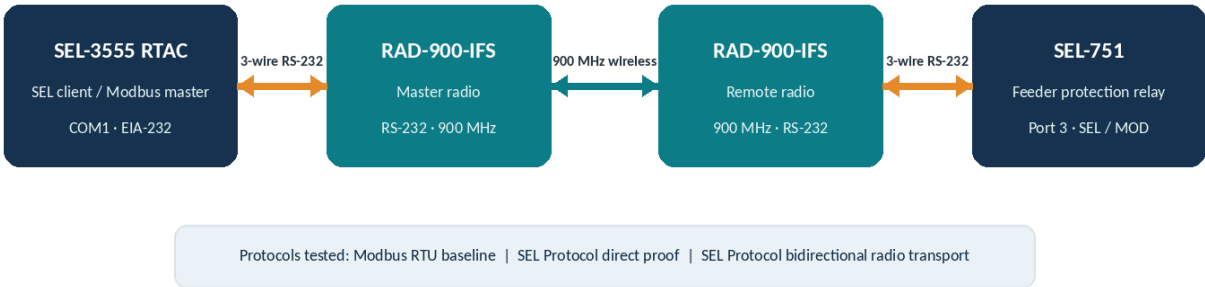


WIRELESS SERIAL COMMUNICATION INTEGRATION

SEL-3555 RTAC ↔ Phoenix RAD-900-IFS ↔ SEL-751 Relay

Bench Test, Troubleshooting Record, and Commissioning Guide

Wireless Serial Communication Architecture



System architecture and protocol test scope

ROLE	WORK TERM	DOCUMENT TYPE	CURRENT STATUS
E&I Reliability Engineering Student	Summer 2026	Engineering project report	Transport proven; final SEL session acceptance pending

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Engineering record designed for technical handover, manager review, and portfolio presentation

OVERVIEW **Project at a Glance**

DIRECT MODBUS PASS Known-good baseline	MODBUS BY RADIO PASS Valid polling; Offline false	DIRECT SEL PASS Relay-native protocol proven	SEL BY RADIO PARTIAL Readable two-way traffic; session open
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The engineering challenge

Create a reliable wireless serial path between an SEL-3555 RTAC and an SEL-751 relay using Phoenix Contact RAD-900-IFS radios, while separating physical-layer problems from protocol, timeout, and login behavior.

The approach

- Prove each protocol directly before adding the wireless radios.
- Use Modbus RTU as a simple request/response baseline, then migrate to SEL Protocol.
- Match every serial parameter across four endpoints and verify the physical interface type.
- Use RTAC Comm Monitor evidence rather than relying on radio LEDs alone.
- Change one variable per test and preserve working configurations as separate projects.

CONCLUSION	Engineering conclusion The radio path is capable of bidirectional EIA-232 transport. Readable SEL-751 text reached the RTAC through the radio pair. The only remaining open item is RTAC SEL-client session acceptance, not basic wiring or RF connectivity.
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SECTION 1. Executive Summary

This project evaluated a wireless industrial serial communications path between an SEL-3555 Real-Time Automation Controller and an SEL-751 feeder protection relay using two Phoenix Contact RAD-900-IFS radios. The work covered direct serial baselines, Modbus RTU, SEL Protocol, EIA-485 and EIA-232 interfaces, radio configuration, DTE/DCE interpretation, voltage and continuity measurements, RTAC communication captures, and final commissioning guidance.

The controlled test sequence produced three fully proven results: direct Modbus RTU communication, Modbus RTU communication through the radio pair, and direct SEL Protocol communication. The radio path also returned readable SEL-751 identification and access text to the RTAC, proving bidirectional wireless serial transport. At the report issue point, the RTAC SEL client still indicated Offline = TRUE through the radio path. Therefore, complete SEL Protocol commissioning is not claimed until the client state remains online and survives a power-cycle recovery test.

PROJECT VALUE

What the project demonstrates

Industrial communications troubleshooting, serial interface design, protocol migration, evidence-based fault isolation, RTAC configuration, relay integration, and disciplined engineering documentation.

Key outcomes

Outcome	Result	Engineering interpretation
Direct Modbus RTU	PASS	Verified RTAC serial polling and relay Modbus configuration.
Modbus RTU through radios	PASS	Proved the radio pair and wiring could carry a valid request and response.
Direct SEL Protocol	PASS	Verified relay-native communication without the wireless link.
SEL traffic through radios	PASS - transport layer	Readable SEL-751 response text reached RTAC RX through the radio pair.
RTAC SEL client online state	OPEN	Requires direct-versus-radio capture comparison and final session tuning.

IMPORTANT

Reporting boundary

Bidirectional readable traffic is strong physical and serial evidence, but it is not equivalent to full application acceptance. Final acceptance requires Offline = FALSE, stable data updates, and successful recovery after power cycling.

SECTION 2. Objectives and Acceptance Criteria

2.1 Project objectives

- Configure a point-to-point wireless serial link using two RAD-900-IFS radios.
- Establish a direct communication baseline before adding the radio pair.
- Verify Modbus RTU and SEL Protocol behavior independently.
- Determine the correct interface type, cable behavior, and signal wiring.
- Create a repeatable troubleshooting and commissioning procedure.
- Produce an engineering handover package with clear evidence and open items.

2.2 Acceptance criteria

Acceptance item	Pass condition	Status
Power and RF link	Both radios remain powered, linked, and free of persistent hardware faults.	PASS
Serial framing	RTAC, both radios, and relay use 9600 / 8 / None / 1 for final SEL testing.	PASS
Direct baseline	SEL Protocol works without radios using the same end-device settings.	PASS
Bidirectional radio transport	RTAC transmits and receives readable SEL-751 text through radios.	PASS
Application online state	RTAC SEL client indicates Offline = FALSE with stable tag updates.	OPEN
Recovery	Communication automatically recovers after power cycling the radio pair.	PENDING

SECTION 3. System Architecture and Signal Flow

Wireless Serial Communication Architecture

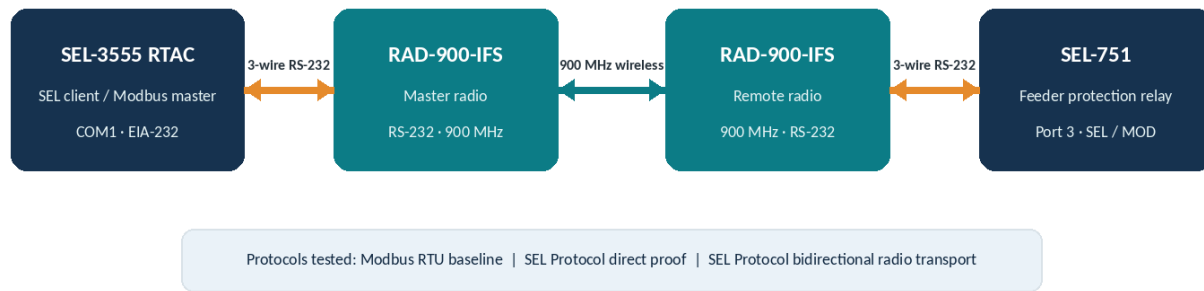


Figure 1 - Final project architecture and tested protocol layers

The RTAC acts as the communications client or master. The SEL-751 acts as the serial server or slave, depending on the selected protocol. The Phoenix radios are transparent serial transport devices: they do not interpret Modbus or SEL Protocol; they carry serial bytes between the two local interfaces.

3.1 Complete request-and-response sequence

1. RTAC generates a request on COM1.
2. The master radio receives the request at its local EIA-232 RX input.
3. The radio pair transports the bytes over the 900 MHz wireless link.
4. The remote radio transmits the request to SEL-751 Port 3.
5. The relay processes the command and generates a response.
6. The response returns through the remote radio, wireless link, master radio, and RTAC RX input.
7. The RTAC protocol driver validates the response and sets the device online or offline.

DIAGNOSTIC RULE

LED interpretation

Master RX and remote TX indicate the command direction. Remote RX and master TX indicate the response direction. LED activity proves byte movement only; protocol captures are required to prove a valid response.

SECTION 4. Hardware, Software, and Electrical Interfaces

Component	Role	Critical project detail
SEL-3555 RTAC	Serial communications client and automation platform	COM1 used for final EIA-232 SEL client test.
SEL-751 relay	Remote protection relay and serial server	Port 3 selected; MOD during baseline and SEL during final test.
2 × RAD-900-IFS	Point-to-point wireless serial bridge	One master and one remote; RS-232 final interface.
SEL-2886 converter	EIA-232 to EIA-485 conversion during alternate tests	Not required in the final pure EIA-232 path.
24 Vdc supplies	Bench power	Common and stable supply arrangements required.
acSELerator RTAC	RTAC configuration and Comm Monitor	Used to configure devices and capture TX/RX evidence.
acSELerator QuickSet	SEL-751 configuration	Used to read Port 3 settings from the physical relay.
Phoenix PSI-CONF	Radio configuration and diagnostics	Used to transfer matching radio serial settings.

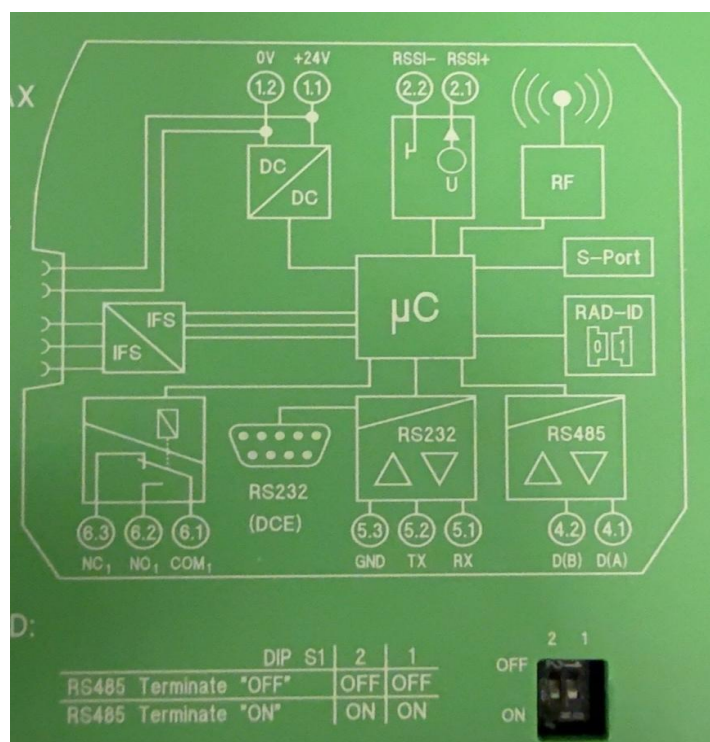


Figure 2 - Replace with a sharp radio nameplate and interface diagram

SECTION 5. Final EIA-232 Wiring

Final EIA-232 Wiring - Signal Direction, Not Wire Colour

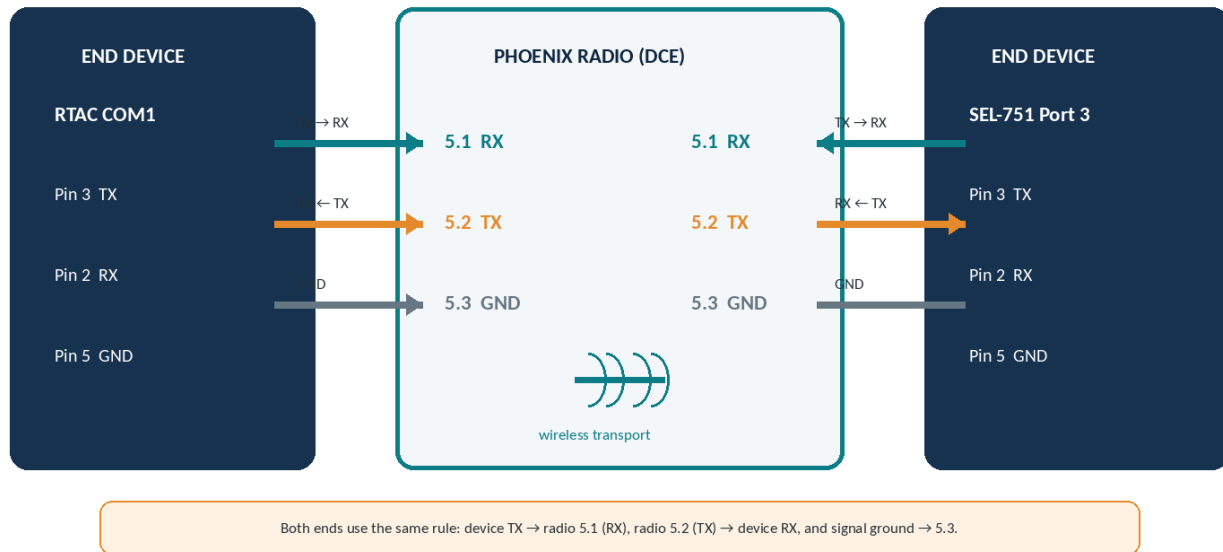


Figure 3 - Final signal-based wiring rule for both ends of the radio link

5.1 Final three-wire connection

End-device signal	Typical DB9 pin used in this bench test	RAD-900-IFS terminal	Function
Device TX	Pin 3	5.1	Data from RTAC or SEL-751 into radio RX.
Device RX	Pin 2	5.2	Data from radio TX into RTAC or SEL-751.
Signal ground	Pin 5	5.3	Common EIA-232 signal reference.

WIRING RULE

DTE/DCE rule

The RAD-900-IFS is marked DCE. A straight-through DB9 cable is appropriate between a DTE-style end device and the radio DB9. When using screw terminals, wire by signal direction: TX to RX, RX to TX, and ground to ground.

5.2 Verification before energization

- De-energize equipment before continuity testing or conductor changes.
- Verify DB9 pin orientation from the connector face; the rear wiring view is mirrored.
- Map conductors by continuity and label them TX, RX, and GND; do not trust cable colour.
- After energization, confirm negative EIA-232 idle voltage from each end-device TX to signal ground.
- Confirm no RTS/CTS or XON/XOFF flow control is enabled for the three-wire connection.

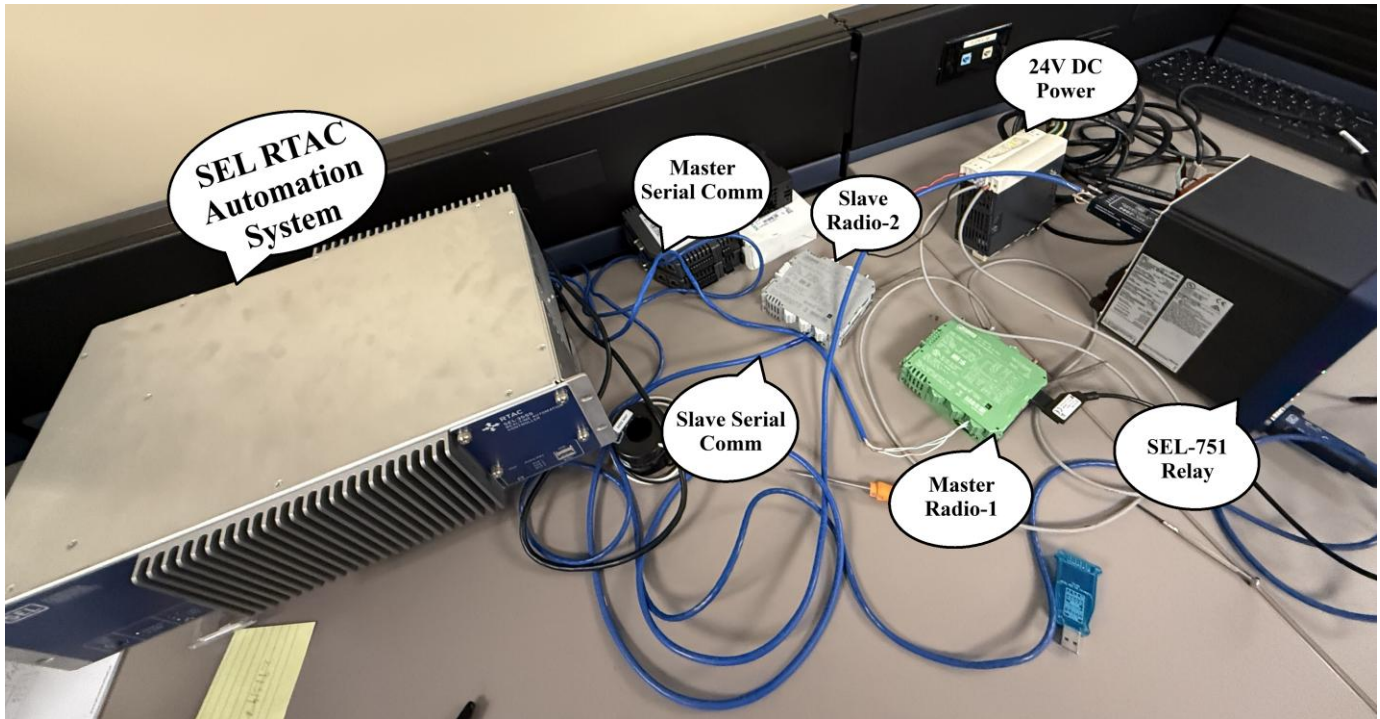


Figure 4 - Replace with the final ferruled and labelled three-conductor wiring photo

SECTION 6. Configuration Baseline

6.1 SEL-3555 RTAC SEL client

Setting	Final test value	Reason
Serial port	COM1	Matches the physical RTAC port used for the EIA-232 bench test.
Port type	EIA232	Matches the radio RS-232 interface.
Baud / data / parity / stop	9600 / 8 / None / 1	Must match all four serial endpoints.
RTS/CTS	False	No hardware handshaking in the three-wire connection.
XON/XOFF	False	No software flow control.
ASCII inactivity timeout	8000 ms test value	Provides margin for radio-path delay.
Binary inactivity timeout	3000 ms test value	Increased from the earlier 500 ms value.
Event / COMTRADE collection	Disabled during first connection test	Reduces unsolicited traffic while commissioning.

6.2 Phoenix RAD-900-IFS pair

Parameter	Required value
Topology	One master and one remote radio, point-to-point
Serial interface	RS-232
Baud rate	9600
Data bits	8
Parity	None

Parameter	Required value
Stop bits	1
Flow control	None
Configuration transfer	Transfer and verify separately on both radios

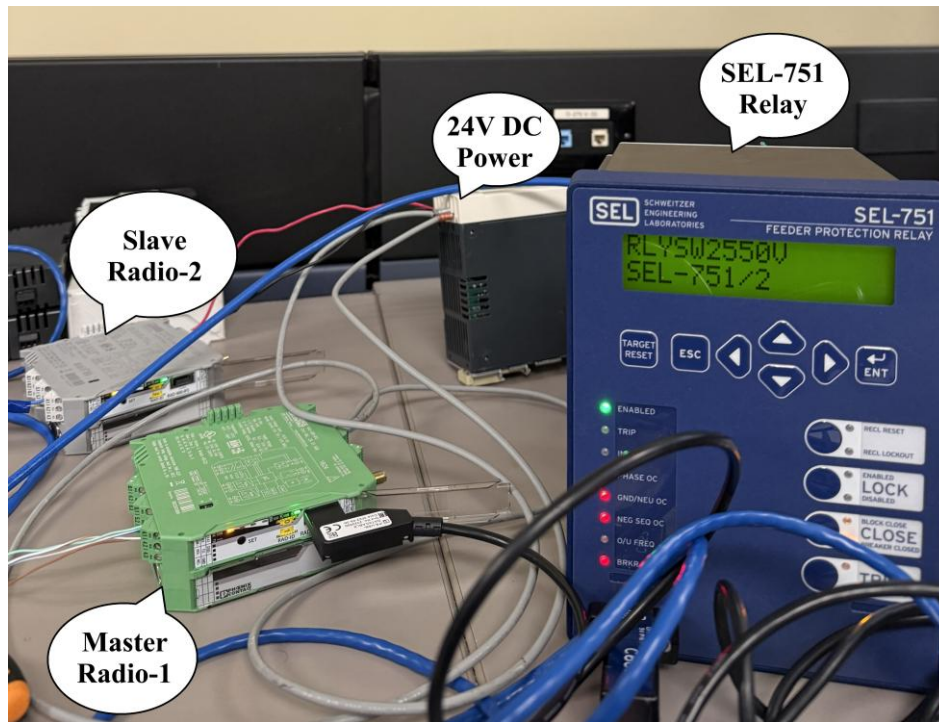


Figure 6 - Replace with the final PSI-CONF settings applied to both radios

6.3 SEL-751 Port 3

Setting	Final SEL Protocol value
Protocol selection	SEL
Maximum access level	2
Data speed	9600 bps
Data bits	8
Parity	N / None
Stop bits	1
RTS/CTS	N / disabled

CONTROL

Configuration control

Save the working Modbus and SEL Protocol configurations as separate RTAC projects. One physical COM port must not be owned by two active protocol clients at the same time.

SECTION 7. Implementation and Controlled Test Method

Controlled Engineering Test Sequence

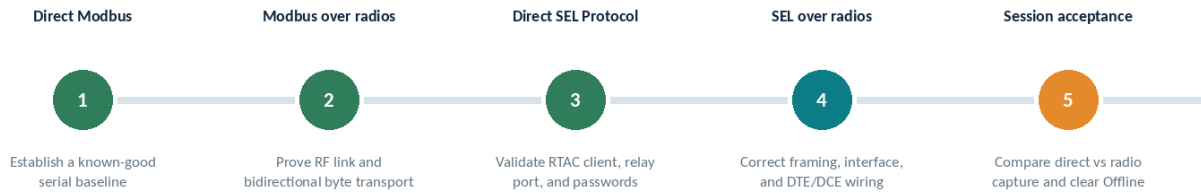


Figure 8 - Test sequence used to isolate hardware, radio, framing, and application layers

7.1 Controlled test discipline

- Capture the baseline before changing settings.
- Prove direct end-device communication before inserting radios.
- Change only one variable per test: protocol, interface type, wiring, framing, or timeout.
- Use Comm Monitor to identify no-transmit, no-response, echo, framing error, readable invalid response, and valid response.
- Record the exact settings and result after each test.
- After success, power cycle the radio pair and confirm automatic recovery.

Test stage	Purpose	Decision enabled
Direct Modbus	Establish a simple known-good request/response baseline.	Confirms relay Modbus map and RTAC serial client.
Modbus through radios	Prove radio hardware, RF link, and serial transport.	Separates radio transport from SEL Protocol issues.
Direct SEL Protocol	Prove relay-native client and Port 3 settings.	Confirms base SEL configuration without radios.
SEL through radios	Validate framing and transparent transport.	Identifies interface, wiring, or timing differences.
Direct-versus-radio capture comparison	Resolve remaining Offline condition.	Separates byte transport from driver session acceptance.

SECTION 8. Modbus RTU Baseline

Modbus RTU was tested first because its request/response behavior is straightforward to capture and validate. During the successful Modbus phase, the SEL-751 used protocol MOD, 9600 baud, 8 data bits, no parity, two stop bits, and slave address 1. The RTAC read holding register address 0, corresponding to the first configured relay Modbus value.

Modbus item	Working test value / result
SEL-751 protocol	MOD
Serial format	9600 / 8 / None / 2 during the working Modbus phase
Slave address	1
RTAC poll	Read holding register, zero-based address 0, quantity 1
Mapped value	MOD_001 = IA magnitude in the relay Modbus map
Radio result	Offline became FALSE and valid responses were recorded

BASELINE

Why this test mattered

A successful Modbus transaction through the radios proved that power, RF communication, the basic serial path, and both directions of transport were capable of carrying a valid industrial protocol.

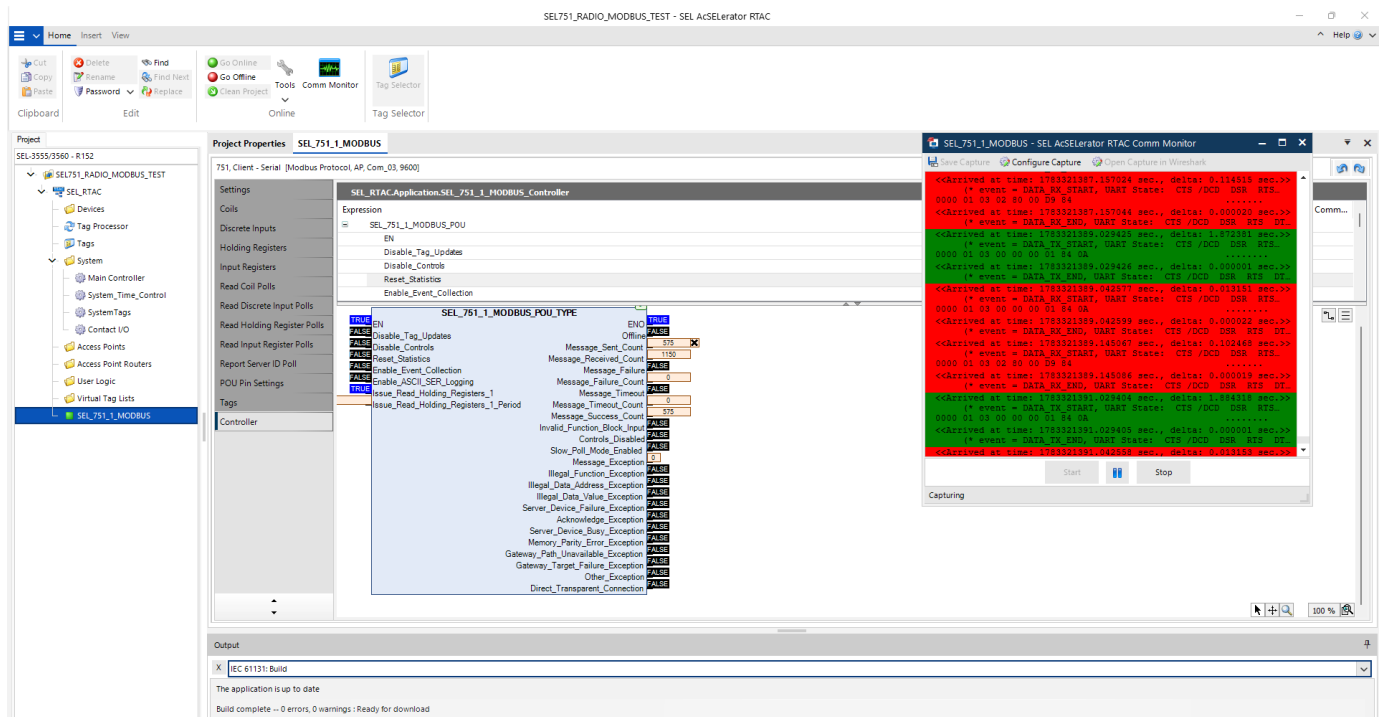


Figure 9 - Replace with the strongest working Modbus-over-radio proof

SECTION 9. SEL Protocol Integration

After the Modbus baseline was proven, a separate RTAC project was created for the SEL client. The SEL-751 Port 3 protocol was changed to SEL, and the complete path was changed to 9600 / 8 / None / 1. The direct SEL connection worked, proving the RTAC SEL client and relay configuration before the radios were inserted.

9.1 Direct proof

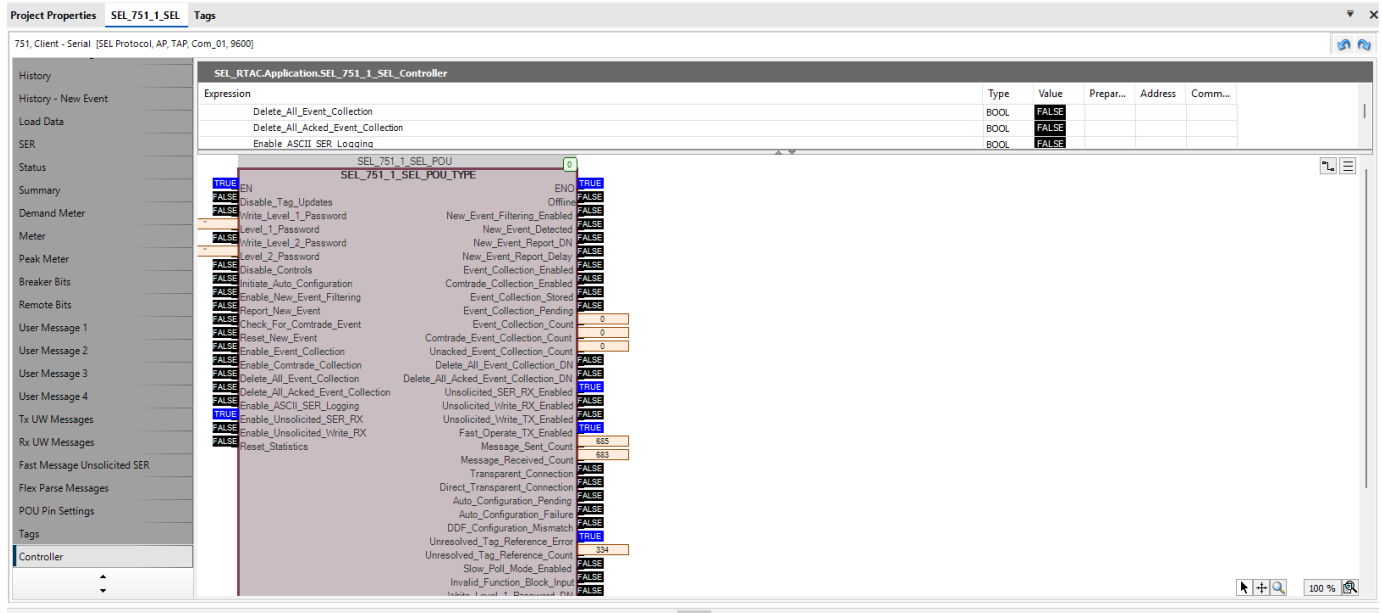


Figure 10 - Replace with the direct SEL Protocol working capture

9.2 Radio-path proof

The final radio-path Comm Monitor capture showed both DATA_TX and DATA_RX events with readable SEL-751 identification and access text. This evidence proves bidirectional serial transport through the radio pair. It does not, by itself, prove that the RTAC SEL driver accepted the complete session.

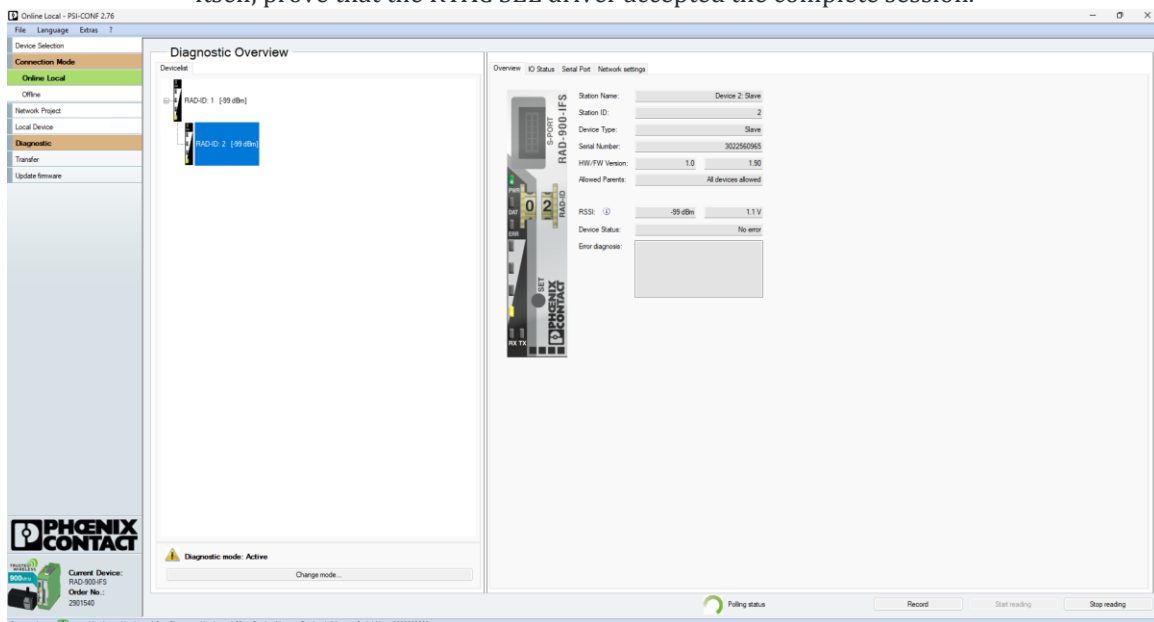


Figure 11 - Replace with the final bidirectional SEL traffic capture

SECTION 10. Troubleshooting and Root-Cause Analysis

Observed symptom	Interpretation	Corrective direction
Comm Monitor blank	Client not transmitting, wrong device selected, project not downloaded, or COM port conflict.	Verify project ownership, download, and online state.
DATA_TX present but master radio RX does not blink	Software is sending but the signal is not reaching radio RX.	Verify EIA232 selection, physical COM port, and TX-to-5.1 wiring.
FRAMING_ERROR	Serial framing mismatch or corrupted byte timing.	Match baud, data bits, parity, and stop bits across all endpoints.
Request echoed back exactly	Local echo, converter echo, or wrong interface behavior.	Disable echo and verify the response is from the relay.
Master RX + remote TX only	Command reaches the relay side, but the response does not return.	Verify SEL-751 TX to remote radio 5.1 and signal ground.
Readable relay text returned	Physical and serial transport are functioning.	Stop rewiring; move to session, password, auto-configuration, and timeout analysis.
Offline remains TRUE with readable data	RTAC driver rejects or times out on the session.	Compare direct and radio captures byte-for-byte.

Common Errors:

- The early SEL test retained two stop bits from the working Modbus configuration, while the relay SEL port used one stop bit. The RTAC captured FRAMING_ERROR. Correcting both radios and both end devices to 9600 / 8 / None / 1 removed the basic framing mismatch.
- The radio nameplate identifies the EIA-232 interface as DCE. A normal straight-through computer cable and an SEL crossover cable are not equivalent. The final wiring was validated by signal direction and voltage rather than cable name or conductor color.
- Continuity confirmed conductor paths but could not prove correct signal direction or valid EIA-232 activity. Negative idle voltage on radio terminal 5.1 relative to 5.3 confirmed that an end-device TX output reached the radio RX input. The strongest proof remained readable protocol data in the RTAC capture.

SECTION 11. Evidence, Current Status, and Open Item

Commissioning Status by Communication Layer

1	Hardware power and RF link Radios powered and wireless link active	PASS
2	EIA-232 electrical path Valid idle levels and TX/RX signal direction established	PASS
3	Serial framing 9600 / 8 / None / 1 matched across all endpoints	PASS
4	Bidirectional relay response Readable SEL-751 text returned to RTAC Comm Monitor	PASS
5	RTAC SEL client session Offline remains TRUE; capture comparison required	OPEN

Figure 13 - Current project status separated by communication layer

11.1 What is proven

- RTAC COM1 transmits EIA-232 data.
- The master radio receives RTAC requests.
- The wireless link delivers requests to the remote radio.
- The SEL-751 receives and processes serial commands.
- The relay generates readable responses.
- The response returns through both radios to RTAC RX.
- The final radio path uses matched 9600 / 8 / None / 1 framing.

11.2 Remaining open item

The RTAC SEL client continued to show Offline = TRUE through the radio path even after readable responses were returned. Because the same SEL client worked directly, the remaining cause is most likely response timing, command/response boundary changes, password/access sequence behavior, auto-configuration state, or radio packetization.

NEXT TEST

Do not change the final wiring without new evidence

Once readable SEL-751 text reaches RTAC RX, repeated wire swapping risks reintroducing a solved physical-layer fault. The next test must compare the working direct capture against the radio-path capture.

11.3 Required next diagnostic test

- Save a complete direct SEL capture from client startup until Offline becomes FALSE.
- Save a radio-path capture over the same interval using identical RTAC and relay settings.
- Compare command order, response text, response delay, missing or duplicated bytes, and carriage-return/line-feed boundaries.
- If the radio capture is byte-complete, review password monitor and auto-configuration outputs.
- If the radio capture is delayed or grouped differently, tune radio frame-end/idle timing and RTAC inactivity timeouts.
- Escalate both captures to SEL technical support if the driver state cannot be resolved.

SECTION 12. Final Commissioning Procedure

Step	Required action	Evidence
1. Hardware	Confirm both radios are powered from 24 Vdc and antennas are installed before transmission.	Power and link LEDs
2. Port ownership	Disconnect QuickSet and disable any other RTAC client using COM1.	One active owner
3. Relay	Read Port 3 from the actual relay: SEL, 9600, 8, N, 1, RTS/CTS disabled.	QuickSet readback
4. RTAC	Set COM1 to EIA232, 9600, 8, None, 1, RTS/CTS false, XON/XOFF false.	RTAC settings
5. Radios	Set both radios to RS-232, 9600, 8, None, 1, no handshake; transfer to each radio.	PSI-CONF readback
6. Wiring	Device TX → 5.1; 5.2 → device RX; signal ground → 5.3 on both sides.	Continuity + voltage
7. Download	Build with zero errors, download to RTAC, and go online.	Build output
8. Monitor	Capture DATA_TX and DATA_RX; verify readable SEL-751 response text.	Comm Monitor
9. Application	Confirm Offline = FALSE and required tags update.	Controller screen
10. Recovery	Power cycle radios and verify automatic communication recovery.	Timed recovery test

Step	Required action	Evidence
11. Handover	Archive projects, settings, captures, wiring drawing, and sign-off.	Handover package

12.1 Final as-built setting sheet

Component	Interface / protocol	Serial format	Notes
SEL-3555 RTAC COM1	EIA-232 / SEL client	9600, 8-N-1	RTS/CTS and XON/XOFF disabled
RAD-900-IFS master	RS-232 / wireless master	9600, 8-N-1	Local 5.1 RX, 5.2 TX, 5.3 GND
RAD-900-IFS remote	Wireless remote / RS-232	9600, 8-N-1	Local 5.1 RX, 5.2 TX, 5.3 GND
SEL-751 Port 3	EIA-232 / SEL Protocol	9600, 8-N-1	Maximum access level 2; RTS/CTS disabled

SECTION 13 13. Safety, Cybersecurity, and Handover

13.1 Safety and reliability

- De-energize bench wiring before continuity checks or conductor changes.
- Install antennas before enabling radio transmission and follow site RF requirements.
- Replace temporary exposed conductors with ferruled, labelled, strain-relieved wiring.
- Use shielded industrial cable and site-approved grounding practices for permanent installation.
- Treat remote control separately from monitoring; perform hazard analysis, interlock verification, and management-of-change review before enabling commands.

13.2 Cybersecurity

- Do not place live relay passwords in the public report or source repository.
- Store credentials using the site-approved credential-management process.
- Wireless encryption does not replace network segmentation, access control, change management, or secure engineering workstations.
- Retain controlled copies of final configurations and record all approved changes.

13.3 Recommended handover package

- Final project report in DOCX and PDF.
- Final RTAC SEL Protocol project export and separate working Modbus baseline project.
- SEL-751 QuickSet settings file read from the physical relay.
- Phoenix PSI-CONF project for both radios.
- As-built EIA-232 wiring diagram with connector orientation.
- Direct and radio-path Comm Monitor captures.
- Completed commissioning checklist and acceptance sign-off.
- Open-item note until the RTAC SEL client remains online after power cycling.

References

Source	Use in this report
Phoenix Contact	RAD-900-IFS installation and product documentation.
Schweitzer Engineering Laboratories	SEL-751 Feeder Protection Relay documentation.
Schweitzer Engineering Laboratories	SEL-3555 Real-Time Automation Controller documentation.
Schweitzer Engineering Laboratories	SEL-2886 EIA-232/EIA-485 Interface Converter instruction manual.
Internal project evidence	RTAC, QuickSet, PSI-CONF, wiring, voltage, radio, and Comm Monitor captures from the July 2026 bench test.